

Diesel Engines
V 595 TE 70 / TE70L**Schedule**
M050589/00E

MTU Friedrichshafen GmbH
88040 Friedrichshafen / Germany
Telephone (07541) 90-0
Telex 734 280-0 mt d
Telefax (07541) 90-3928

Low operating and maintenance costs as well as operational reliability and availability depend on maintenance and servicing being carried out in compliance with our specifications and instructions.

The overall system, of which the engine is an integral part, must be maintained in such a way as to ensure trouble-free engine operation at all times. For this purpose always:

- ensure that sufficient fuel is available,
- ensure that the combustion air is dry and clean,
- use only dry, clean compressed air,
- use only clean, filtered raw water.

Moreover, it is essential that:

- maintenance tasks be completed by trained personnel,
- suitable tools be used,
- genuine spare parts and fluids and lubricants as per the current MTU Fluids and Lubricants Specification A001061 be used.

Our Product Support Service will always be available should assistance be required.

Preventive maintenance instructions.

- Special care should be taken to keep the machinery plant in a clean and serviceable condition at all times to facilitate early detection of possible leaks and prevent subsequent damage.
- Protect rubber and synthetic components from oil and fuel, never treat with organic detergents. Wipe with a dry cloth only.
- Always replace all seals and gaskets.

MTU Maintenance Concept

The maintenance concept for MTU products is a preventive maintenance concept and features the maintenance levels W1 to W6 as outlined below.

Preventive maintenance facilitates advance planning and ensures a high degree of equipment availability.

Main features of the maintenance levels:

- W1: Operational checks
- W2, W3 and W4: Periodic maintenance tasks to be performed during out-of-service periods without the need for engine disassembly.
- W5: This level requires partial engine disassembly.
- W6: Major overhaul. This level requires complete engine disassembly.

The time intervals according to which the W2 to W6 Maintenance Levels and the relevant checks and tasks involved are to be completed are based on operational experience. They have been determined so as to ensure correct engine operation until the next scheduled maintenance level.

Use of a maintenance predictor can mean changes to the operating hours given in the Maintenance Frequency Chart.

The oil-change intervals must be determined depending on the results of the lube-oil analysis. When changing the oil, the engine-oil filter must also be replaced.

Specific operational conditions may require modification of the Maintenance Schedule.

NOTE:

The codes on the following pages refer to the groups and subgroups of the Task Descriptions in Section G of the Engine Operation Manual.

This Maintenance Schedule may refer to parts which are not installed on your engine, these may be disregarded.

Application Group

1B: Fast vessels with high load factors

Load profile

Load referenced to fuel stop power	%:	100	85	< 15
Time element %	%:	3	82	15

Maintenance Frequency Chart

Maintenance Level W1:	Operational checks, daily	X
W2:	Operating hours	500
	Limit, months	6
W3:	Operating hours	1 000
	Limit, year	1
W4:	Operating hours	4 000
	Limit, year(s)	2
W5:	Operating hours	8 000
	Limit, year(s)	6
W6:	Operating hours	24 000
	Limit, year(s)	18

Maintenance Task Schedule

Code No.	One-time operations after the first 50 operating hours. – With a new engine, after W5 maintenance level, or after W6 major overhaul	
G00.008	Attachments	Check tightness of securing screws and nuts
G05.505	Valve gear	Check valve clearances
G08.305	Fuel duplex filter	Replace filter elements
G08.305.1	Fuel prefilter	Replace filter elements
G18.002	Engine oil	Take sample and analyze
G18.305	Engine oil filter	Operate filter handle several times
G20.610.1	Engine coolant system (HT circuit)	Remove strainer before thermostat
G20.615.1	Engine coolant system (LT circuit)	Remove strainer before thermostat

Code No.	Maintenance Level W1: – Operational checks	
G00.005	Engine operation	Check running noises, Check engine and external pipework for leaks, Check operating conditions with Operational Data Recorder, note deviations
G08.305.1	Fuel prefilter	Drain water and contaminants Check differential pressure
G12.105	Air filter	Check contamination indicator
G14.000	Exhaust system	Check exhaust gas color Drain condensate (if drain valve provided)
G18.001	Engine oil	Check level
G20.000	Engine coolant	Check level
G20.360	Charge-air system	Check air-supply drain for water discharge and obstructions
G50.000	ECS / LOP	Carry out lamp test